Address		Name
Mem.	I/O	
$20	$00	TWBR
$21	$01	TWSR
$22	$02	TWAR
$23	$03	TWDR
$24	$04	ADCL
$25	$05	ADCH
$26	$06	ADCSRA
$27	$07	ADMUX
$28	$08	ACSR
$29	$09	UBRRL
$2A	$0A	UCSRB
$2B	$0B	UCSRA
$2C	$0C	UDR
$2D	$0D	SPCR
$2E	$0E	SPSR
$2F	$0F	SPDR
$30	$10	PIND
$31	$11	DDRD
$32	$12	PORTD
$33	$13	PINC
$34	$14	DDRC
$35	$15	PORTC

Address		Name
Mem.	I/O	
$36	$16	PINB
$37	$17	DDRB
$38	$18	PORTB
$39	$19	PINA
$3A	$1A	DDRA
$3B	$1B	PORTA
$3C	$1C	EECR
$3D	$1D	EEDR
$3E	$1E	EEARL
$3F	$1F	EEARH
$40	$20	UBRRC
		UBRRH
$41	$21	WDTCR
$42	$22	ASSR
$43	$23	OCR2
$44	$24	TCNT2
$45	$25	TCCR2
$46	$26	ICR1L
$47	$27	ICR1H
$48	$28	OCR1BL
$49	$29	OCR1BH
$4A	$2A	OCR1AL

Address		Name
Mem.	I/O	
$4B	$2B	OCR1AH
$4C	$2C	TCNT1L
$4D	$2D	TCNT1H
$4E	$2E	TCCR1B
$4F	$2F	TCCR1A
$50	$30	SFIOR
$51	$31	OCDR
		OSCCAL
$52	$32	TCNT0
$53	$33	TCCR0
$54	$34	MCUCSR
$55	$35	MCUCR
$56	$36	TWCR
$57	$37	SPMCR
$58	$38	TIFR
$59	$39	TIMSK
$5A	$3A	GIFR
$5B	$3B	GICR
$5C	$3C	OCR0
$5D	$3D	SPL
$5E	$3E	SPH
$5F	$3F	SREG

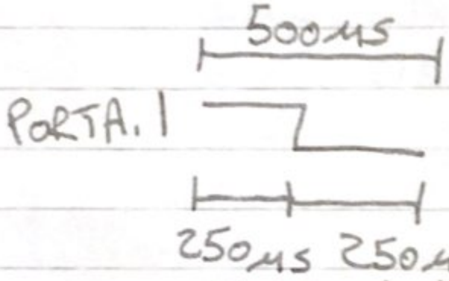
Note: Although memory address $20-$5F is set aside for I /O registers (SFR) we can access them as I /O locations with addresses starting at $00.

Figure 2-7. I/O Registers of the ATmega32 and Their Data Memory Address Locations

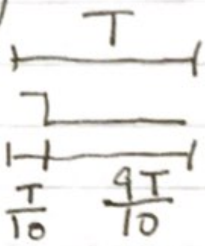
$S_{50} = S_{10} S_{10}$

Generate a 2KHz square wave on PortA, bit1. Assume $f_{osc} = 16 MHz$

$$2KHz \Rightarrow T = \frac{1}{f} = \frac{1}{2(10^3)} = 0.5(10^{-3})$$
$$= 5(10^{-4})$$
$$= 500(10^{-6})$$
$$= 500\,\mu s$$



50% duty cycle



10% duty cycle

$t_{delay} = 250\mu s$

$t_{delay} = t_{MC}\, n$

t_{MC} = time per MC (machine cycle)

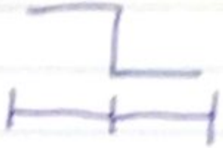
n = # MCs to generate time delay

$$t_{MC} = \left(\frac{1 \text{ clock cyc}}{1 MC}\right)\left(\frac{1 s}{16(10^6) cc}\right)$$
$$= 0.0625 \mu s / MC$$

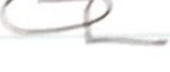
$$n = \frac{t_{delay}}{t_{MC}}$$

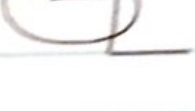
$$n = \frac{250 \mu s}{0.0625 \mu s / MC} = 4000 \text{ MCs}$$

4000 MCs ⇒ 250μs



```
; Main program
.ORG 0
LDI R16,0xFF
OUT DDRA,R16; making all bits PORTA
                output (only need PA.1)
              ; SBI DDRA, 1 (PA.1
                             output)
X MCs
 2    loop: SBI PORTA,1 ; PORTA.1 = 1
 3          RCALL Delay ;
 1          NOP
 1          NOP
 2          CBI PORTA,1 ; PORTA.1=0
 3          RCALL Delay ;
 2          RJMP loop
```





$\overset{SBI}{2} + \overset{RCALL}{3} + \text{Delay} + \overset{NOP}{1} + \overset{NOP}{1}$

=

$\overset{CBI}{2} + \overset{RCALL}{3} + \text{Delay} + \overset{RJMP}{2}$

4004 ⇒ 4011MC

0-255 8bits

4000MCs ⇒ 200 (20)
↑ ↑
<255 inside
Loop1

```
          ; Delay Routine
#MCs        .ORG 0x100
1    Delay: LDI R18, 200
     Loop1: NOP
              :
Y             :     } 17 NOPs
              :
            NOP
1           DEC R18
2/1         BRNE Loop1
4           RET
```

$$\overset{\text{LDI}}{1} + (y + \overset{\text{DEC}}{1} + \overset{\text{BRNE}}{2})200 \overset{\text{BRNE}}{-1} \overset{\text{RET}}{+4}$$

$y+1+2=20$

$y=17$ MCs

4004 MCs ⇒ close enough to 4000

Similar to previous
loop example without
timing considerations

```
#MCs
1     Delay:  LDI R18, 200
1     Loop2: LDI R19, 20
1     Loop1: DEC R19
2/1          BRNE Loop1
1            DEC R18
2/1          BRNE Loop2
4            RET
```

$1_{\text{LDI}} + [(1_{\text{DEC}}+2_{\text{BRNE}})20-1_{\text{BRNE}} + 1_{\text{LDI}}+1_{\text{DEC}}+2_{\text{BRNE}}]200- 1_{\text{BRNE}} + 4_{\text{RET}}$
$= 1+[(3)20-1+1+1+2]200-1+4$
$= 1+[63]200-1+4 = 12604$ MCs

tDelay = 12604MCs(1cc/1MC) (1s/16(10^6)cc)
= 788 uS (much greater than desired half period time delay)
Therefore, need to be more exact with Delay instruction sequence to get closer to 4000MCs.

Replace 17 NOPs with a loop (loop2)

loop2: $\overset{LDI}{1} + (\overset{DEC}{1} + \overset{BRNE}{2})x \overset{BRNE}{-1} + \overset{Nop}{1} + \overset{Nop}{1} = 17$

$3x = 17 \Rightarrow$ yuck!!

$2 + 3x = 17 \Rightarrow 3x = 15 \Rightarrow x = 5$

```
 .ORG 0x100
Delay: LDI R18,200

Loop1: LDI R24,5
[redacted] NOP
       NOP
Loop2: DEC R24
       BRNE Loop2
       DEC R18 ; decrement loop1 counter
       BRNE Loop1
       RET
```